

Literature-Implied Status Note for arXiv:0802.3520

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Status. `literature_implied_answer` for the two-sided lattice compactness conjecture in arXiv:0802.3520. The supporting source is a recent preprint, so human verification is recommended before treating the question as settled.

Source/open-problem metadata. Michael Cwikel, “Complex interpolation of compact operators mapping into lattice couples,” arXiv:0802.3520, posted 2008-02-24. In the subsection “Two sided interpolation of compactness,” near the end of the paper (local TeX source lines 1005–1021), Cwikel writes that it was not even known whether the stronger two-sided endpoint condition

$$T : A_j \xrightarrow{c} B_j, \quad j = 0, 1,$$

implies

$$T : [A_0, A_1]_\theta \xrightarrow{c} [B_0, B_1]_\theta, \quad 0 < \theta < 1.$$

He conjectures this implication for arbitrary Banach couples $\vec{A} = (A_0, A_1)$ whenever $\vec{B} = (B_0, B_1)$ is an arbitrary lattice couple, with no Fatou-property or σ -order-continuity assumptions.

Supporting-answer metadata. Evgeniy Pustylnik, “A new approach to interpolation of compact linear operators,” arXiv:2605.00530, posted 2026-05-01. The paper states that compactness is interpolated automatically whenever one has standard interpolation of bounded operators, first for two-sided compactness; one-sided compactness needs an additional condition. It explicitly includes the complex interpolation functor $\mathcal{F}_{[\theta]}$, $0 < \theta < 1$, noting the estimate $W(\alpha, \beta) = \alpha^{1-\theta} \beta^\theta$ in the one-sided case. In the final section, item 1 says that the same proof applies to two-sided compactness of embedding operators without the one-sided inequality. Item 2 then passes from embeddings to arbitrary operators by factoring an operator through its image spaces.

Identification. Cwikel’s conjecture is a two-sided compactness statement for the complex interpolation functor, restricted to arbitrary lattice couples on the range side. Pustylnik’s statement is broader: it is formulated for general Banach couples and interpolation functors satisfying standard bounded-operator interpolation, with complex interpolation explicitly included. For an arbitrary operator T that is compact $A_j \rightarrow B_j$ for both endpoints, the image-space factorization $T = JS$ makes $S : A_j \rightarrow Y_j$ bounded and makes the embeddings $J : Y_j \hookrightarrow B_j$ compact at both endpoints. Combining the final-section two-sided embedding remark with this same factorization gives compactness of $T : \mathcal{F}(A_0, A_1) \rightarrow \mathcal{F}(B_0, B_1)$. Taking $\mathcal{F} = [\cdot, \cdot]_\theta$ yields the implication conjectured in arXiv:0802.3520, if the supporting preprint is accepted.

Provenance limitation. This is an agent-identified implication rather than an explicit answer announced by the supporting authors. The checked source text of arXiv:2605.00530 does not appear to cite arXiv:0802.3520 or Cwikel’s two-sided lattice-couple conjecture. Since arXiv:2605.00530 is a very recent preprint claiming a broad resolution of a long-standing compactness interpolation problem, human review should verify the proof and the embedding-to-operator passage before marking the conjecture definitively closed.

Search evidence. The run indexes were searched for 0802.3520, complex interpolation, compact operators, lattice couples, two-sided compactness, and arXiv:2605.00530. No prior exact packet for arXiv:0802.3520 was found. A related existing packet for arXiv:1406.3842 already records that arXiv:2605.00530 appears to answer the classical one-sided complex compactness problem; the present packet records the separate implication for Cwikel’s two-sided lattice-couple conjecture.

References

- [1] M. Cwikel, “Complex interpolation of compact operators mapping into lattice couples,” arXiv:0802.3520, 2008.
- [2] E. Pustylnik, “A new approach to interpolation of compact linear operators,” arXiv:2605.00530, 2026.